

Harder Trigonometry Exam Questions 1

Total - 62 Marks (1 hour, 36 minutes)

1. (Caringbah 2025 Q10c)

Solve

4

$$\cos^2 x + \cos^2(2x) + \cos^2(3x) = 1, 0 \leq x \leq 2\pi$$

2. (Sydney Girls 2021 Q4d)

Prove that: $\tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) = 2 \sec 2\theta$

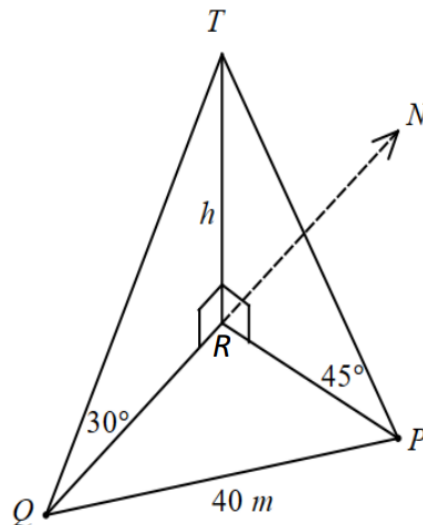
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3. (Abbotsleigh 2025 Q11e)

Solve $\sqrt{2} \cos\left(x + \frac{\pi}{4}\right) + \sqrt{2} \cos\left(x - \frac{\pi}{4}\right) = 1$ for $-\pi \leq x \leq \pi$.

2

4. (Girraween 2025 Trials Q29) A vertical tower of height h metres stands on horizontal ground. From a point P on the ground due east of the tower the angle of elevation of the top of the tower is 45° . From a point Q on the ground due south of the tower the angle of elevation of the top of the tower is 30° . If the distance PQ is 40 metres, find the exact height of the tower.



5. (Girraween 2024 Q4 a, b, d & e)

(a) Using the expansion of $\cos(A + B)$, find the exact value of $\sec 105^\circ$

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(b)

i. Prove that $\tan \theta = \sqrt{\frac{1 - \cos 2\theta}{1 + \cos 2\theta}}$

2

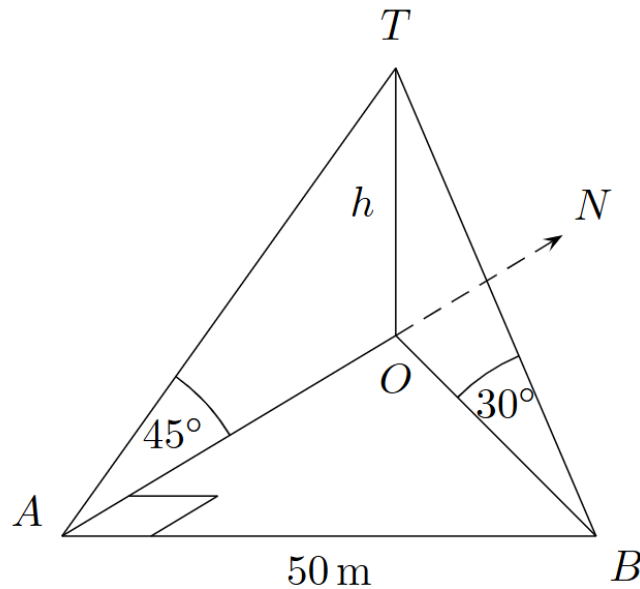
ii. Hence show the exact value of $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$

2

(d) Show that $\sin(A - 2B) \cos(A + 2B) = \frac{1}{2}(\sin 2A - \sin 4B)$ **2** Note: Sine is an odd function and Cosine is an even function.

(e) Prove the identity $\frac{2 \sin^3 A + 2 \cos^3 A}{\sin A + \cos A} = 2 - \sin 2A$

2



6. (Normanhurst Boys 2024 Y11 Half-Yearly Q16) A person is standing at point A , which is due south of a tower OT that is h metres tall. The person walks 50 metres east to point B . The angle of elevation at points A and B to the top of the tower are 45° and 30° respectively.

- Express OA and OB in terms of h . 2
- Show that $h = 25\sqrt{2}$ metres. 2
- Find the bearing of B from the base of the tower O , correct to the nearest minute. 2

7. (Glenwood 2023 Q12b)

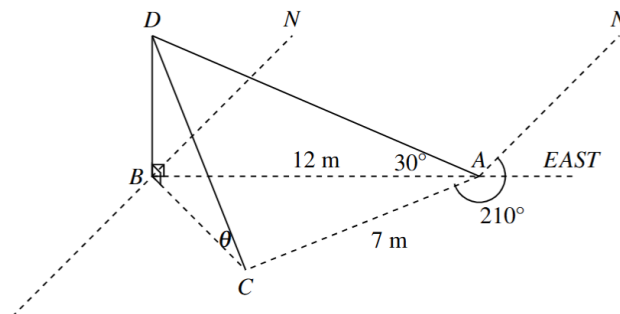
Express $9 \sin x + 40 \cos x$ in the form $A \sin(x + \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$, and 4

hence or otherwise solve $9 \sin x + 40 \cos x = 6$ for $0 \leq x \leq \pi$.

(Leave your answer for x and α in radians, correct to three significant figures).

8. (Penrith 2023 Preliminary Q30)

A flagpole, BD , sits on flat level ground. It is supported by two cables, AD and CD , as shown below. The cable AD is inclined at 30° to the horizontal and is tied to the floor 12 metres from the foot of the flagpole. The second cable is tied to the floor at point C , 7 metres from A and on a bearing of 210° from A .

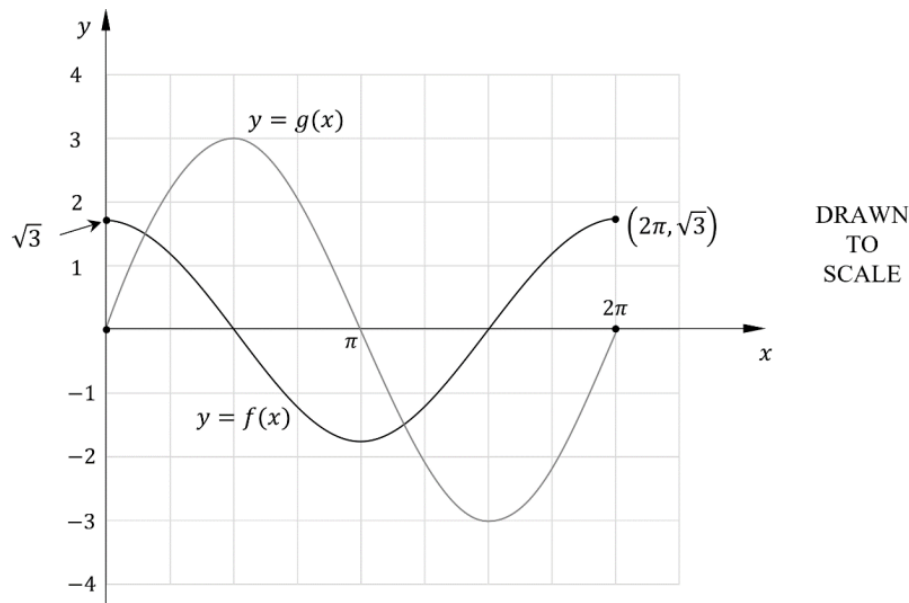


- Show that $\angle BAC = 60^\circ$. 1
- Find the exact height of the flagpole, BD . 1
- Find the exact distance B to C and hence, find θ , correct to the nearest degree 3

9. (Caringbah 2023 Q13d)

The graphs of the trigonometric curves $y = f(x)$ and $y = g(x)$ are shown below.

3



Find the equation of the curve $y = f(x) + g(x)$, expressed in the form $y = R \cos(x - \alpha)$, where α is in radians.

10. (Baulkham Hills 2024 Q11c)

(i) Show that $\cot \theta - \cot 2\theta = \operatorname{cosec} 2\theta$.

2

(ii) Hence find the exact value of $\cot 15^\circ$.

1

11. (Merewether 2025 Q11f)

Solve $\cos 2\theta = \cos^2 \theta + \cos \theta + 1$ for $0 \leq \theta \leq 2\pi$

2

12. (Ascham 2023 Q13a)

i) Show that $\tan 2x + \tan x = \frac{\sin 3x}{\cos 2x \cos x}$.

2

ii) Hence solve the equation $\tan 2x + \tan x = 0$ for $0 \leq x \leq \frac{\pi}{2}$.

2

13. (Sydney Girls 2020 Q4c)

Suppose that x is a real number and that m and n are positive integers.

(i) Express $\sin mx \sin \frac{x}{2}$ as a difference of cosine functions.

1

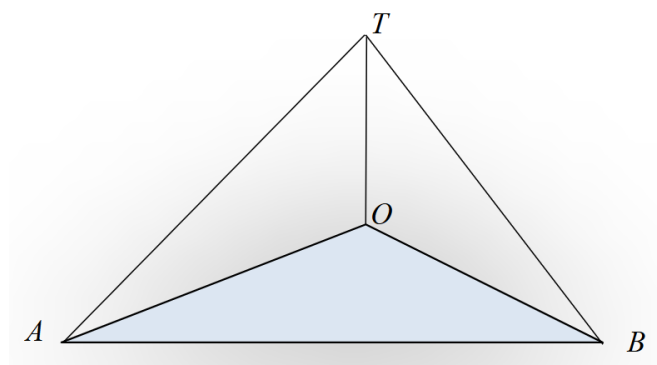
(ii) Hence, show that

2

$$\sin x + \sin 2x + \sin 3x + \cdots + \sin nx = \frac{\cos \frac{x}{2} - \cos \left(n + \frac{1}{2}\right)x}{2 \sin \frac{x}{2}}.$$

14. (Hornsby Girls 2023 Y11 Half-Yearly Q11b)

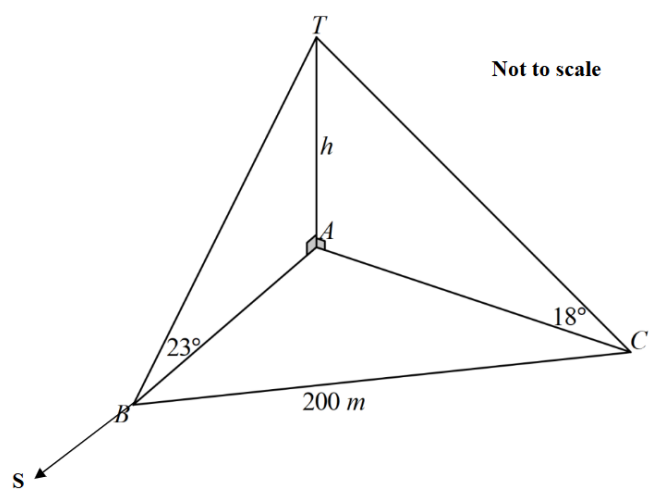
Alex walks along a straight road on level ground. At one point, A , a tower is noticed on a bearing of 053° , with an angle of elevation of 21° . After Alex walks a further 230 m to a point B , the tower is on a bearing of 342° with an angle of elevation of 26° . Let the base of the tower be the point O and the top of the tower be the point T .



(i) State the size of $\angle AOB$. 1

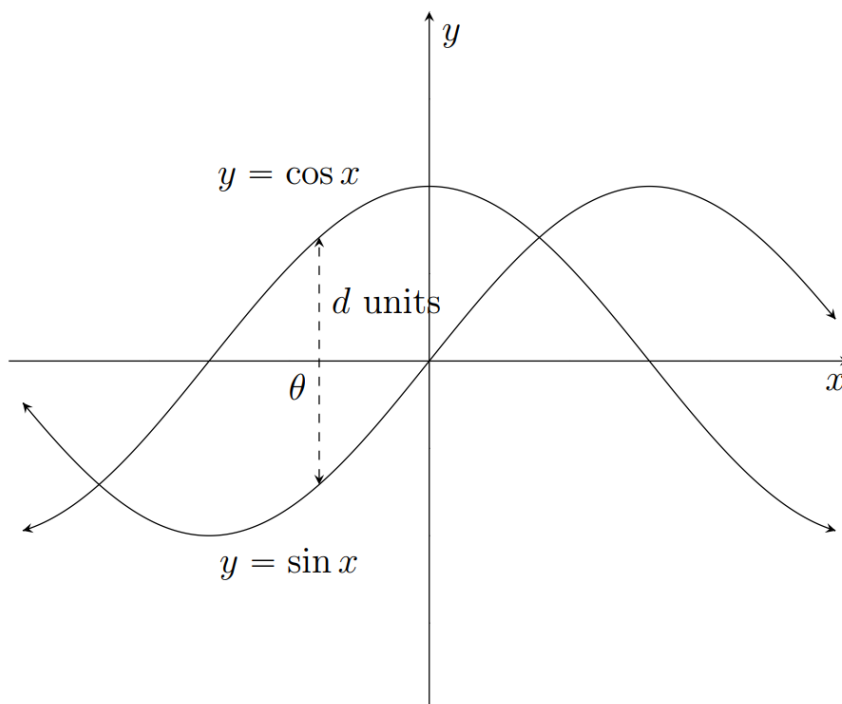
(ii) Find the height of the tower correct to the nearest metre. 3

15. (Hornsby Girls 2024 Y11 Half-Yearly Q9d) TA is a vertical tower of height h metres. B and C are points at the same level as A , the foot of the tower, and $BC = 200$ m. The point B is due south of the tower and C is on a bearing of $120^\circ T$ from the base of the tower. The angles of elevation to the top of the tower from B and C are 23° and 18° , respectively. Find the height of the tower, correct to one decimal place. 4



16. (Sydney Girls 2025 Q13d)

The graphs of $y = \cos x$ and $y = \sin x$ are sketched below.



The dashed line indicates the greatest vertical distance, d units, between the two curves.

The x -coordinate of the dashed line is θ .

(i) Find d , giving reasons for your answer.

1

(ii) Find θ , giving reasons for your answer.

1